25 Particle and Photon Orbits in the Schwarzschild Geometry

We have now accumulated all the ingredients to address an issue of fundamental interest in general relativity, namely the study of planetary orbits and lightrays in the gravitational field of the sun, i.e. the properties of timelike and null geodesics of the Schwarzschild geometry.

We shall see that, by making good use of the symmetries of the problem, we can reduce the (initially somewhat complicated looking) set of coupled 2nd order geodesic equations to a single 1st order differential equation in one variable, analogous to that for a one-dimensional particle moving in a particular ("effective") potential. Solutions to this equation can then readily be discussed qualitatively and also quantitatively (analytically).

In order to be able to read this section as a concrete ("real-life", so to speak) application and illustration of the general formalism for geodesics introduced in sections 2 and 3 (without knowledge of all the intervening material on tensor analysis and the Einstein equations, say), this section starts with a brief summary of the key features and main properties of the Schwarzschild metric that we will make use of in the following and that were established in detail in the previous section 24.

25.1 Schwarzschild Metric: Summary of its Key Properties

In standard Schwarzschild coordinates (t, r, θ, ϕ), the Schwarzschild metric is

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \quad , \tag{25.1}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the line element of the standard (rotationally invariant) metric on the 2-sphere S^2, and $f(r)$ is the function

$$f(r) = 1 - \frac{2m}{r} \quad , \tag{25.2}$$

with m an integration constant (with the dimension of length).

This metric has the following origin / interpretation / properties:

1. As shown in section 24.2, via a suitable choice of coordinates, any metric that is static and spherically symmetric (and thus describes a time-independent spherically symmetric gravitational field) can be put into the form

   $$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \quad , \tag{25.3}$$

   for some (positive) functions $A(r)$ and $B(r)$.

   Provided that $A(r) \to 1$ and $B(r) \to 1$ (sufficiently rapidly) as $r \to \infty$, this metric is asymptotically Minkowskian (usually referred to as *asymptotically flat*).

An anymptotically flat metric can be considered to describe the gravitational field of an isolated/localised object in space.

In particular, therefore, the Schwarzschild metric is static, spherically symmetric and asymptotically flat.

2. What singles out the Schwarzschild metric among all such metrics is the crucial fact (established in section 24.3) that

   The Schwarzschild metric is the unique static, spherically symmetric and asymptotically flat solution of the vacuum Einstein field equations!

   [This statement can be sharpened somewhat: the assumption that the metric is static turns out to be unnecessary - see the discussion of Birkhoff's theorem in section 24.6.]

   Here "vacuum Einstein field equations" refers to the field equations for the metric in the absence of matter or in matter-free regions of space-time. Thus the physical interpretation of the Schwarzschild metric is that it describes the gravitational field in the exterior region $r > r_0$ of a static spherically symmetric star of radius r_0.

3. Comparison of the Schwarzschild

$$g_{00} = -\left(1 - \frac{2m}{r}\right) \tag{25.4}$$

   with the Newtonian limit, where one has (cf. the discussion in section 3.3)

$$g_{00} = -\left(1 + \frac{2\phi}{c^2}\right) \tag{25.5}$$

   shows that this physical interpretation is perfectly compatible with the Newtonian limit, where the appropriate Newtonian potential ϕ for a star with mass M has the form

$$\phi(r) = -\frac{G_N M}{r} \quad . \tag{25.6}$$

   In particular, the physical interpretation of the integration constant m appearing in the Schwarzschild solution is that it is related to the mass M by

$$m = \frac{G_N M}{c^2} \quad , \tag{25.7}$$

   The Schwarzschild metric is thus the extension of the Newtonian $1/r$-potential, the unique exact spherically symmetric solution of the Newtonian "vacuum" field equation

$$\Delta\phi(r) = 0 \quad \text{for} \quad r > r_0 \tag{25.8}$$

   vanishing as $r \to \infty$, to an exact solution of the general relativistic vacuum field equations for the metric.

4. Here is what we can say about the coordinates:

   - The Schwarzschild coordinates are adapted to the symmetries of the metric in the sense that invariance under time translations $t \to t + b$ and rotations of the angular coordinates (θ, ϕ) is manifest, or as manifest as possible.
   - The coordinates (θ, ϕ) have the standard interpretation and range as coordinates on the 2-sphere.
   - Even though r is not a measure of proper radial distance, it does have a clean geometric interpretation, namely that r is such that spheres of radius r are those that have area $4\pi r^2$.
   - A priori the range of r is limited by the requirement $r > r_0$ (because the Schwarzschild metric only describes the gravitational field to the exterior of the star). One might also be concerned about the behaviour of the metric as $r \to 2m$ but, as recalled below, for standard astrophysical objects like planets or stars one has $r_0 \gg 2m$, so that the condition $r > r_0$ automatically excludes this potentially dangerous region.
   - Since the metric is independent of t, the range of the coordinate t can be taken to be $(-\infty, +\infty)$. Up to affine transformations $t \to at + b$, t is also characterised by the fact that the time-translation symmetry is realised by translations $t \to t + b$ of the coordinate t (and not some more complicated transformation of the coordinates).
   - The physical interpretation of t is that it is the proper time of a Minkowskian observer "at infinity" (where $f(r) \to 1$) and that, up to a constant conversion factor $\sqrt{f(r)} < 1$ it is the proper time of observers hovering at fixed values of the spatial coordinates (r, θ, ϕ).

   Since symmetries are manifest, and the coordinates have a simple interpretation, these coordinates are ideally suited for studying the geodesics, i.e. the motion of particles and light, in the gravitational field of an object like the sun, with $r_0 \gg 2m$, and we will embark on this below.

   However, for other purposes other coordinates may be more useful or even necessary and can provide additional insight. We will look at this in quite some detail in subsequent sections.

5. As shown in (24.40), the apparently problematic behaviour of the Schwarzschild metric as $r \to 2m$ is irrelevant for applications to the solar system (or the description of the gravitational field of other standard benign astrophysical objects), since for such objects m is much less than the radius of the object, $m \ll r_0$ (while the Schwarzschild metric only applies to the exterior region $r > r_0$).

   E.g. for the earth m is of the order of a centimeter, while for an object with a mass approximately that of the sun it is of the order of a couple of kilometers.

However, we will take a closer look at what happens at (and beyond) $r = 2m$ in section 26 and subsequent sections, when exploring the physics of black holes.

25.2 Symmetries and Effective Potential for the Kepler Problem

To set the stage for the subsequent discussion of the motion of particles (and light) in the Schwarzschild geometry, let us very briefly recall how one goes about this in the corresponding (Kepler) problem in Newtonian mechanics, namely the motion of particles in the potential (25.6).

The crucial point here is to make judicious use of the symmetries of the problem to reduce the problem of the 3-dimensional motion of a particle in the spherically symmetric potential $V(r)$ to that of a 1-dimensional radial motion in a suitable *effective potential*. Then the problem has been reduced to that of solving a single ordinary differential equation (and this can be done in closed form for the Kepler problem), and even without knowledge of the exact solutions the qualitative behaviour of the orbits in the Schwarzschild geometry can essentially be determined "by inspection" from drawing the effective potential.

The steps are the following:

1. Because of time-translation invariance, the Newtonian energy E_N of a particle moving in an arbitary time-independent potential $V(\vec{x})$ is conserved. This energy is the sum of the kinetic energy and the potential energy. Passing straightaway to spherical coordinates (r, θ, ϕ), and just looking at a particle of unit mass (since we already know that in the case of the Kepler problem the mass of the particle will drop out of all equations anyway) this energy is

$$E_N = \frac{1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2\right) + V(r,\theta,\phi) \quad , \tag{25.9}$$

   where (in this Newtonian context) an overdot refers to an ordinary time-derivative, $\dot{r} = dr/dt$ etc.

2. When the potential is purely radial (spherically symmetric), $V = V(r)$, the problem is invariant under spatial rotations, and correspondingly angular momentum $\vec{L}$ is conserved. Thus motion takes place in the plane orthogonal to $\vec{L}$. For any choice of $\vec{L}$ one can choose the coordinate system such that $\vec{L}$ points into the x^3-direction or, equivalently, such that the motion is restricted to the equatorial plane $\theta = \pi/2$,

$$\theta = \pi/2 \quad \Rightarrow \quad \vec{L} = (0,0,L) \quad , \quad L = r^2\dot{\phi} \ . \tag{25.10}$$

3. Plugging this back into the energy, one finds

$$E_N = \frac{1}{2}\dot{r}^2 + \frac{L^2}{2r^2} + V(r) \equiv \frac{1}{2}\dot{r}^2 + V_{eff}(r) \quad , \tag{25.11}$$

where $V_{eff}(r)$ is known as the *effective potential*. It governs the radial motion of the particle in the potential $V(r)$ and differs from $V(r)$ by the usual centrifugal / angular momentum barrier term (which acts as a potential for the corresponding centrifugal "pseudo-force").

Thus the symmetries have allowed us to reduce the original set of 3 2nd-order differential equations for the coordinates $\vec{x} = \vec{x}(t)$ to a single ordinary 1st-order differential equation for $r = r(t)$,

$$\frac{dr}{dt} = \pm\sqrt{2E_N - 2V_{eff}(r)} \ . \tag{25.12}$$

4. To obtain an equation for the *orbit*, i.e. for $r = r(\phi)$, one can use

$$\frac{dr}{d\phi} = \frac{\dot{r}}{\dot{\phi}} = \pm\frac{r^2}{L}\sqrt{2E_N - 2V_{eff}(r)} \ . \tag{25.13}$$

As is well known, for the Kepler problem, with $V(r) \sim 1/r$, this equation can be solved in closed form in terms of conical sections (this is recalled at the beginning of section 25.8).

25.3 Symmetries and Effective Potential for Schwarzschild Geodesics

Our aim is now to try to develop a similarly efficient strategy to deal with the corresponding general relativistic problem, i.e. for describing the solutions of the geodesic equation in the Schwarzschild geometry.

A convenient starting point in general for discussing geodesics is, as I stressed repeatedly, the Lagrangian $\mathcal{L} = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$. For the Schwarzschild metric this is

$$\mathcal{L} = -(1-2m/r)\dot{t}^2 + (1-2m/r)^{-1}\dot{r}^2 + r^2(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) \ , \tag{25.14}$$

where $2m = 2MG_N/c^2$.

Rather than writing down and solving the (second order) geodesic equations, we will make use of the conserved quantities associated with the continuous symmetries of the Lagrangian (or the metric). After all, conserved quantities correspond to first integrals of the equations of motion and if there are a sufficient number of them (there are) we can directly reduce the second order differential equations to first order equations.

1. Spherical Symmetry and Conserved Angular Momentum

   Since the gravitational field is spherically symmetric, or isotropic, there is conservation of angular momentum. Thus, exactly as in the Newtonian problem, the orbits of the particles or planets are planar. Without loss of generality, we can choose our coordinates in such a way that this plane is the equatorial plane $\theta = \pi/2$, so in particular $\dot{\theta} = 0$ at all times.

In case this is not obvious (even though it should be), here are two explicit ways to establish this:

- One can certainly choose one's coordinates in such a way that at some initial time $\tau = \tau_0$ one has $\theta(\tau_0) = \pi/2$ and that the angular velocity $\dot\theta(\tau_0) = 0$. It then follows from the Euler-Lagrange equations of motion for $\theta(\tau)$ that they are solved at all times by $\theta(\tau) = \pi/2$, $\dot\theta(\tau) = 0$.
- If you already know about Killing vectors (which made their first appearance in this context in sections 3.2), you may find it somewhat more insightful to use the conserved angular momenta

$$L_{(a)} = g_{\alpha\beta}\dot x^\alpha V^\beta_{(a)} \tag{25.15}$$

associated to the Killing vectors $V_{(a)}$ (9.55),

$$\begin{aligned} L_{(1)} &= r^2\left(-\cos\phi\,\dot\theta + \cot\theta\sin\phi(\sin^2\theta\,\dot\phi)\right)\\ L_{(2)} &= r^2\left(+\sin\phi\,\dot\theta + \cot\theta\cos\phi(\sin^2\theta\,\dot\phi)\right)\\ L_{(3)} &= r^2\sin^2\theta\,\dot\phi\ , \end{aligned} \tag{25.16}$$

and to use spherical symmetry to rotate the $L_{(a)}$ into the form

$$(L_{(1)}, L_{(2)}, L_{(3)}) = (0, 0, L)\ . \tag{25.17}$$

Using the explicit expressions (25.16), it is straightforward to see that $L_{(1)} = L_{(2)} = 0$ implies $\theta = \pi/2$, $\dot\theta = 0$.

Either way, we have fixed the direction of $\vec L$, the motion is now restricted to the equatorial plane, with

$$\theta = \pi/2 \quad\Rightarrow\quad \vec L = (0, 0, L)\ , \tag{25.18}$$

and the residual Lagrangian to deal with is

$$\mathcal{L} = -(1 - 2m/r)\dot t^2 + (1 - 2m/r)^{-1}\dot r^2 + r^2\dot\phi^2\ . \tag{25.19}$$

The magnitude L (or z-component of $\vec L$) of the angular momentum (per unit rest mass) is now the conserved quantity

$$L = r^2\dot\phi \tag{25.20}$$

associated to the cyclic variable ϕ of the residual Lagrangian,

$$\frac{\partial\mathcal{L}}{\partial\phi} = 0 \Rightarrow \frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\dot\phi} = 0\ . \tag{25.21}$$

2. Time Translation Invariance and Conserved Energy

   The Lagrangian (both the original Lagrangian and the reduced Lagrangian) is invariant under translations of t, which is also a cyclic variable,

$$\frac{\partial \mathcal{L}}{\partial t} = 0 \Rightarrow \frac{d}{d\tau}\frac{\partial \mathcal{L}}{\partial \dot{t}} = 0 \ . \tag{25.22}$$

   This gives rise to the conserved energy (with a conventional choice of normalisation)

$$E = (1 - 2m/r)\dot{t} \ . \tag{25.23}$$

   Calling L the angular momentum (per unit rest mass) required no further justification, but let me pause to explain in what sense E is an energy (per unit rest mass).

   - On the one hand, it is the conserved quantity (3.7) associated to time-translation invariance. As such, it certainly deserves to be called the energy.
   - It is moreover true that for a particle at infinity ($r \to \infty$) E is just the special relativistic energy $E = \gamma(v_\infty)c^2$, with $\gamma(v) = (1 - v^2/c^2)^{-1/2}$ the usual relativistic γ-factor, and v_∞ the coordinate velocity dr/dt at infinity. This can be seen in two ways. First of all, for a particle that reaches $r = \infty$, the constant E can be determined by evaluating it at $r = \infty$. It thus follows from the definition of E that

$$E = \dot{t}_\infty \ . \tag{25.24}$$

     In Special Relativity, the relation between proper and coordinate time is given by (setting $c = 1$ again)

$$d\tau = \sqrt{1 - v^2}dt \quad \Rightarrow \quad \dot{t} = \gamma(v) \ , \tag{25.25}$$

     suggesting the identification

$$E = \gamma(v_\infty) \qquad (E = \gamma(v_\infty)c^2 \quad \text{if} \quad c \neq 1) \tag{25.26}$$

     Another argument for this identification will be given below, once we have introduced the effective potential.

3. τ-Translation Invariance and the Conserved Lagrangian $\mathcal{L}$

   As we have seen in section 2.5 (and again in section 5.8), there is also always one more integral of the geodesic equation, namely $\mathcal{L}$ itself,

$$\frac{d}{d\tau}\mathcal{L} = 2g_{\mu\nu}\dot{x}^\mu D_\tau \dot{x}^\nu = 0 \tag{25.27}$$

   (this can be interpreted as the conserved "Hamiltonian" associated with the invariance of $\mathcal{L}$ under translations of the affine parameter). Thus we set

$$\mathcal{L} = \epsilon \ , \tag{25.28}$$

where $\epsilon = -1$ for timelike geodesics and $\epsilon = 0$ for null geodesics. We thus have

$$-(1-2m/r)\dot{t}^2 + (1-2m/r)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = \epsilon \ , \tag{25.29}$$

Putting everything together, we can now express $\dot{t}$ and $\dot{\phi}$ in terms of the conserved quantities E and L to obtain a first order differential equation for r alone, namely

$$-(1-2m/r)^{-1}E^2 + (1-2m/r)^{-1}\dot{r}^2 + \frac{L^2}{r^2} = \epsilon \ . \tag{25.30}$$

Multiplying by $(1-2m/r)/2$ and rearranging the terms, one obtains

$$\frac{E^2+\epsilon}{2} = \frac{\dot{r}^2}{2} + \epsilon\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \ . \tag{25.31}$$

Now this equation is of the familiar Newtonian form

$$E_{eff} = \frac{\dot{r}^2}{2} + V_{eff}(r) \ , \tag{25.32}$$

with

$$\begin{aligned} E_{eff} &= \frac{E^2+\epsilon}{2} \\ V_{eff}(r) &= \epsilon\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \ , \end{aligned} \tag{25.33}$$

describing the energy conservation for the 1-dimensional motion in an effective potential. Except for $t \to \tau$, this is exactly the same as the Newtonian equation of motion in a potential

$$V(r) = \epsilon\frac{m}{r} - \frac{mL^2}{r^3} \ , \tag{25.34}$$

the effective angular momentum term $L^2/r^2 = r^2\dot{\phi}^2$ arising, as usual, from the change to polar coordinates. As in the corresponding Newtonian one-dimensional (radial) problem, the qualitative behaviour of the orbits in the Schwarzschild geometry can thus essentially be determined "by inspection".

Given that in principle we started off with the four coupled non-linear geodesic differential equations, this is an enormous and enormously useful simplification, and the main result of this section.

REMARKS:

1. In particular, for $\epsilon = -1$, the general relativistic motion (as a function of τ) is exactly the same as the Newtonian motion (as a function of t) in the potential

$$\epsilon = -1 \ \Rightarrow \ V(r) = -\frac{m}{r} - \frac{mL^2}{r^3} \ . \tag{25.35}$$

The first term is just the ordinary Newtonian potential, so the second term is apparently a general relativistic correction. We will later on treat this as a perturbation but note that the above is an exact result, not an approximation (so, for example, there are no higher order corrections proportional to higher powers of m/r). We expect observable consequences of this general relativistic correction because many properties of the Newtonian orbits (Kepler's laws) depend sensitively on the fact that the Newtonian potential is precisely $\sim 1/r$.

2. In order to see that the new 3rd term really is a (very) small correction to the Newtonian effective potential, note first of all that, restoring a factor of c^2 in the first (Newtonian) potential term, we can write the effective potential as

$$V_{eff}(r) = -\frac{G_N M}{r} + \frac{L^2}{2r^2} - \frac{m}{r}\frac{L^2}{r^2} \quad . \tag{25.36}$$

For the planetary orbits, there is a balance between the attractive 1st and repulsive 2nd term, so these two terms are of the same order of magnitude. The 3rd term is then smaller than these by a factor (m/r). Since r (the radial coordinate of the planet) is much larger than r_0 (the radius of the sun) and, as we had already seen before, for standard astrophysical objects like the sun the radius r_0 is much larger than the Schwarzschild radius $2m$, we have

$$r \gg r_0 \gg m \quad \Rightarrow \quad \frac{m}{r} \ll 1 \quad . \tag{25.37}$$

Thus, the general relativistic corrction term $\sim mL^2/r^3$ indeed provided a very small (but important) correction to the Newtonian potential.

3. Looking at the equation for $\epsilon = -1$,

$$\tfrac{1}{2}\dot{r}^2 + V_{eff}(r) = \tfrac{1}{2}(E^2 - 1) \tag{25.38}$$

and noting that $V_{eff}(r) \to 0$ for $r \to \infty$, we can read off that for a particle that reaches $r = \infty$ we have the relation

$$\dot{r}_\infty^2 = E^2 - 1 \quad . \tag{25.39}$$

This implies, in particular, that for such (scattering) trajectories one necessarily has $E \geq 1$, with $E = 1$ corresponding to a particle initially or finally at rest at infinity. For $E > 1$ the coordinate velocity at infinity can be computed from

$$v_\infty^2 = \frac{\dot{r}_\infty^2}{\dot{t}_\infty^2} \quad . \tag{25.40}$$

Using (25.24) and (25.39), one finds

$$v_\infty^2 = \frac{E^2 - 1}{E^2} \quad \Leftrightarrow \quad E = (1 - v_\infty^2)^{-1/2} \quad , \tag{25.41}$$

thus confirming the result claimed in (25.26).

4. From the effective potential we can deduce the radial geodesic equation of motion, namely

$$\epsilon = -1 \quad \Rightarrow \quad \ddot{r} = -V'_{eff}(r) = -\frac{m}{r^2} + (r - 3m)\frac{L^2}{r^4} \ . \tag{25.42}$$

5. For null geodesics, $\epsilon = 0$, on the other hand, the Newtonian part of the potential is zero, as one might expect for massless particles, but in General Relativity a photon with $L \neq 0$ feels a non-trivial potential

$$\begin{aligned} \epsilon = 0 \quad \Rightarrow \quad V(r) &= -\frac{mL^2}{r^3} \\ V_{eff}(r) &= f(r)\frac{L^2}{2r^2} \ . \end{aligned} \tag{25.43}$$

6. It will also turn out to be useful to have the full radial part of the null geodesic equation at our disposal. As in the timelike case above, this can simply be read off from the effective potential and one finds

$$\epsilon = 0 \quad \Rightarrow \quad \ddot{r} = -V'_{eff}(r) = \frac{L^2}{r^4}(r - 3m) \ . \tag{25.44}$$

Obviously something potentially interesting is happening at $r = 3m$, provided the star is sufficiently small so that its radius $r_0 < 3m$ of course - see section 25.10). Also note that this differs from its timelike counterpart (25.42) precisely by the absence of the Newtonian term $-m/r^2$.

7. Finally, we note that the success of the above analysis relied only on the symmetries of the metric, not on the fact that the particular metric we were looking at satisfies the vacuum Einstein equations. It is indeed reasonably straightforward to generalise the preceding analysis to arbitrary static spherically symmetric metrics. As mentioned before, among these general static spherically symmeric metrics the class of metrics (24.75)

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2 \tag{25.45}$$

is of particular importance and interest. For these metrics one finds that the geodesic equation can still be reduced to the effective potential form (25.32), with E_{eff} precisely as in (25.33), and V_{eff} now given by

$$V_{eff}(r) = -\epsilon\phi(r) + \frac{L^2}{2r^2} + \phi(r)\frac{L^2}{r^2} \ , \tag{25.46}$$

where we have written $f(r)$ in terms of the corresponding "Newtonian" potential $\phi(r)$ as

$$f(r) = 1 + 2\phi(r) \ . \tag{25.47}$$

For the Schwarzschild metric one has $\phi(r) = -m/r$ and (25.46) reduces to (25.33).

25.4 Caveat: Effective Potential versus Routhian

Because this tends to cause some confusion, not just among students but also occasionally in the literature, before turning to a more detailed analysis of the consequences of the equations we have just derived, I want to insert a cautionary remark, a *caveat*, regarding the above derivation of the effective potential equation:

Namely, starting from the Lagrangian $\mathcal{L}$ (25.14), considerations about spherical symmetry led us to choose $\theta = \pi/2$, and imposing this condition in the Lagrangian we were led to the reduced Lagrangian in (25.19). This is legitimate, since such *holonomic* constraints can always be inserted into the Lagrangian itself (this is one of the main virtues and advantages of the Lagrangian formalism).

Subsequently we used the conserved quantities E and L associated to the cyclic variables t and ϕ in order to eliminate from the Lagrangian the quantities $\dot{t}$ and $\dot{\phi}$. Here one needs to be more careful. In general such non-holonomic constraints like $r^2\dot{\phi} = L$ cannot be inserted into the Lagrangian to obtain a (reduced) Lagrangian from which the equations of motion for the remaining variables (here r) can be obtained as the Euler-Lagrange equations. One correct way to do this is to pass to what is known as the *Routhian*, a partial Legendre transform of the Lagrangian on the cyclic variables.

As a simple example, consider the Lagrangian of a (unit mass) free particle in 2 dimensions, expressed in polar coordinates. Its Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}(\dot{r}^2 + r^2\dot{\phi}^2) \ . \tag{25.48}$$

The variable ϕ is cyclic, leading to the conserved angular momentum

$$p_\phi = r^2\dot{\phi} = L \ , \tag{25.49}$$

and the equation of motion for r is the Euler-Lagrange equation

$$\ddot{r} = r\dot{\phi}^2 \ . \tag{25.50}$$

It is legitimate to eliminate $\dot{\phi}$ in this equation, leading to the centrifugal force equation

$$\ddot{r} = L^2/r^3 \ . \tag{25.51}$$

Had one eliminated $\dot{\phi}$ in the Lagrangian instead, one would have found the reduced Lagrangian and resulting Euler-Lagrange equations

$$\mathcal{L} \to \tfrac{1}{2}(\dot{r}^2 + L^2/r^2) \quad \Rightarrow \quad \ddot{r} = -L^2/r^3 \qquad \text{(wrong!)} \ . \tag{25.52}$$

However, even though you might have gained that impression, this is of course not what we were doing in our derivation of the radial effective potential equation. There we were using this elimination in conjunction with the condition that $\mathcal{L}$ is constant on

solutions, to obtain a first-order equation for r (the effective potential equation), and this *is* legitimate. This is a 1st integral of the 2nd order radial equation, the 2nd order equations for r follow from this simply by differentiating with respect to proper time τ, and *not* by treating $\dot{r}^2/2 + V_{eff}(r)$ as a Lagrangian and looking at its Euler-Lagrange equations.

And finally, just to illustrate the claim about the Routhian in the above example, the partial Legendre transform with respect to the cyclic variable ϕ of $\mathcal{L}$ is (expressing $\dot{\phi}$ in terms of p_ϕ)

$$\mathcal{R} = \mathcal{L} - p_\phi \dot{\phi} = \tfrac{1}{2}(\dot{r}^2 - p_\phi^2/r^2) \to \tfrac{1}{2}(\dot{r}^2 - L^2/r^2) \tag{25.53}$$

(note the sign flip compared with the "wrong" Lagrangian above). This is the correct reduced Lagrangian, whose Euler-Lagrange equations give rise to the correct radial equation of motion.

25.5 Equation for the Shape of the Orbit

Typically, one is primarily interested in the shape of an orbit, that is in the radius r as a function of ϕ, $r = r(\phi)$, rather than in the dependence of, say, r on some extraterrestrial's proper time τ. In this case, the above mentioned difference between t (in the Newtonian theory) and τ (here) is irrelevant: In the Newtonian theory one uses $L = r^2 d\phi/dt$ to express t as a function of ϕ, $t = t(\phi)$ to obtain $r(\phi)$ from $r(t)$. In General Relativity, one uses the analogous equation $L = r^2 d\phi/d\tau$ to express τ as a function of ϕ, $\tau = \tau(\phi)$. Hence the shapes of the General Relativity orbits are precisely the shapes of the Newtonian orbits in the potential (25.34). Thus we can use the standard methods of Classical Mechanics to discuss these general relativistic orbits and of course this simplifies matters considerably.

To obtain r as a function of ϕ we proceed as indicated above. Thus we use

$$\left(\frac{dr}{d\phi}\right)^2 = \frac{\dot{r}^2}{\dot{\phi}^2} \tag{25.54}$$

to combine (25.32),

$$\dot{r}^2 = 2E_{eff} - 2V_{eff}(r) \ , \tag{25.55}$$

and (25.20),

$$\dot{\phi}^2 = \frac{L^2}{r^4} \tag{25.56}$$

into

$$\frac{r'^2}{r^4} L^2 = 2E_{eff} - 2V_{eff}(r) \tag{25.57}$$

where a prime denotes a ϕ-derivative.

In the examples to be discussed below, we will be interested in the angle $\Delta\phi$ swept out by the object in question (a planet or a photon) as it travels along its trajectory between

the farthest distance r_2 from the star (sun) ($r_2 = \infty$ for scattering trajectories) and the position of closest approach to the star r_1 (the perihelion or, more generally, if we are not talking about our own solar system, periastron), and back again,

$$\Delta\phi = 2\int_{r_1}^{r_2} \frac{d\phi}{dr} dr \quad . \tag{25.58}$$

In the Newtonian case, these integrals can be evaluated in closed form. With the general relativistic correction term, however, these are elliptic integrals which cannot be expressed in closed form. A perturbative evaluation of these integrals (treating the exact general relativistic correction as a small perturbation) also turns out to be somewhat delicate since e.g. the limits of integration depend on the perturbation.

It is somewhat simpler to deal with this correction term not at the level of the solution (integral) but at the level of the corresponding differential equation. As in the Kepler problem, it is convenient to make the change of variables

$$u = \frac{1}{r} \qquad u' = -\frac{r'}{r^2} \quad . \tag{25.59}$$

Then (25.57) becomes

$$u'^2 = L^{-2}(2E_{eff} - 2V_{eff}(r)) \quad . \tag{25.60}$$

Upon inserting the explicit expression for the effective potential, this becomes

$$u'^2 + u^2 = \frac{E^2 + \epsilon}{L^2} - \frac{2\epsilon m}{L^2} u + 2mu^3 \quad . \tag{25.61}$$

This can be used to obtain an equation for $d\phi(u)/du = u'^{-1}$, leading to

$$\Delta\phi = 2\int_{u_2}^{u_1} \frac{d\phi}{du} du \quad . \tag{25.62}$$

Differentiating (25.61) once more, one finds

$$u'(u'' + u) = u'(-\frac{\epsilon m}{L^2} + 3mu^2) \quad . \tag{25.63}$$

Thus either $u' = 0$, which corresponds to a circular orbit of constant radius (i.e. the solution is $u(\phi) = u_0$ or $r(\phi) = r_0$)), and this is not only a trivial but also an irrelevant solution since neither the planets nor the photons of interest to us travel on circular orbits, or

$$u'' + u = -\frac{\epsilon m}{L^2} + 3mu^2 \quad . \tag{25.64}$$

The unperturbed ("Newtonian") equation is the linear equation obtained by dropping the last term, and its solutions $u_0(\phi)$,

$$u_0'' + u_0 = -\frac{\epsilon m}{L^2} \quad , \tag{25.65}$$

give the familiar conical sections for $\epsilon = -1$ (cf. section 25.8) and straight lines for $\epsilon = 0$ (section 25.11). Treating the last term as a small perturbation, one can then expand the solution u as

$$u = u_0 + u_1 \tag{25.66}$$

with u_1 small, leading to the linear equation

$$u_1'' + u_1 = 3mu_0^2 \tag{25.67}$$

for the perturbation u_1, with u_0 providing the source term (or external force on the harmonic oscillator).

Equations (25.65) and (25.67) are the equations that we will study below to determine the perihelion shift and the bending of light by a star. In the latter case, which is a bit simpler, I will also sketch two other derivations of the result, based on different perturbative evaluations of the elliptic integral.

25.6 Timelike Geodesics

We will first try to gain a qualitative understanding of the behaviour of geodesics in the effective potential

$$V_{eff}(r) = -\frac{m}{r} + \frac{L^2}{2r^2} - \frac{mL^2}{r^3} \quad . \tag{25.68}$$

The standard way to do this is to plot this potential as a function of r for various values of the parameters L and m. The basic properties of $V_{eff}(r)$ are the following:

1. Asymptotically, i.e. for $r \to \infty$, the potential tends to the Newtonian potential,

$$V_{eff}(r) \stackrel{r\to\infty}{\longrightarrow} -\frac{m}{r} \quad . \tag{25.69}$$

2. At the Schwarzschild radius $r_s = 2m$, nothing special happens and the potential is completely regular there,

$$V_{eff}(r = 2m) = -\frac{1}{2} \quad . \tag{25.70}$$

   For the discussion of planetary orbits in the solar system we can safely assume that the radius of the sun is much larger than its Schwarzschild radius, $r_0 \gg r_s$, but the above shows that even for these highly compact objects with $r_0 < r_s$ geodesics are perfectly regular as one approaches r_s. Of course the particular numerical value of $V_{eff}(r = 2m)$ has no special significance because $V(r)$ can always be shifted by a constant.

3. The extrema of the potential, i.e. the points at which $dV_{eff}/dr = 0$, are at

$$mr^2 - L^2 r + 3mL^2 = 0 \quad \Rightarrow \quad r_\pm = (L^2/2m)[1 \pm \sqrt{1 - 12(m/L)^2}] \quad , \tag{25.71}$$

   and the potential has a maximum at r_- and a local minimum at r_+. Thus there are qualitative differences in the shapes of the orbits between $L/m < \sqrt{12}$ and $L/m > \sqrt{12}$.

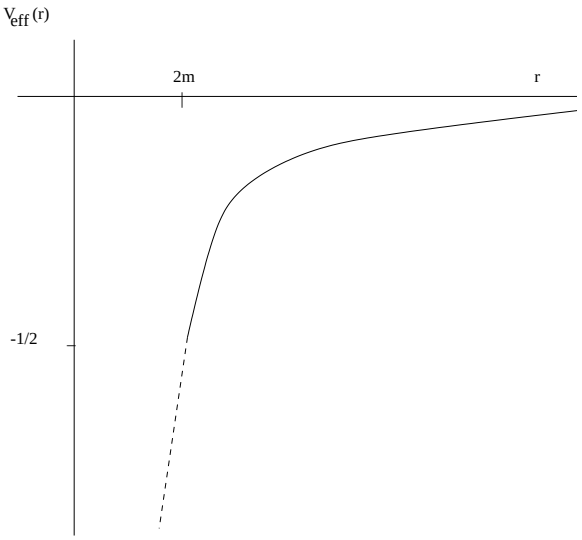


Figure 12: Effective potential for a massive particle with $L/m < \sqrt{12}$. The extrapolation to values of $r < 2m$ has been indicated by a dashed line.

Let us discuss these two cases in turn. When $L/m < \sqrt{12}$, then there are no critical points and the potential looks approximately like that in Figure 12. Note that for the time being we should be careful with extrapolating to values of r with $r < 2m$ because we know that the Schwarzschild metric has a coordinate singularity there. However, the picture (in fact the entire effective potential) turns out to be correct also for $r < 2m$.

From this picture we can read off that there are no bounded orbits for these values of the parameters. Any inward bound particle with $L < \sqrt{12}m$ will continue to fall inwards (provided that it moves on a geodesic). This should be contrasted with the Newtonian situation in which for any $L \neq 0$ there is always the centrifugal barrier reflecting incoming particles since the repulsive term $L^2/2r^2$ will dominate over the attractive $-m/r$ for small values of r. In General Relativity, on the other hand, it is the attractive term $-mL^2/r^3$ that dominates for small r.

Fortunately for the stability of the solar system, the situation is qualitatively quite different for sufficiently large values of the angular momentum, namely $L > \sqrt{12}m$ (see Figure 13).

In that case, there is a minimum and a maximum of the potential. The critical radii correspond to exactly circular orbits, unstable at r_- (on top of the potential) and stable at $r_+ > r_-$ (the minimum of the potential). We will briefly discuss these separately in section 25.7 below.

For given L, for sufficiently large values of E_{eff} a particle will fall all the way down the potential. For $E_{eff} < 0$, there are bound orbits which are not circular and which range between the radii r_1 and r_2, the turning points at which $\dot{r} = 0$ and therefore

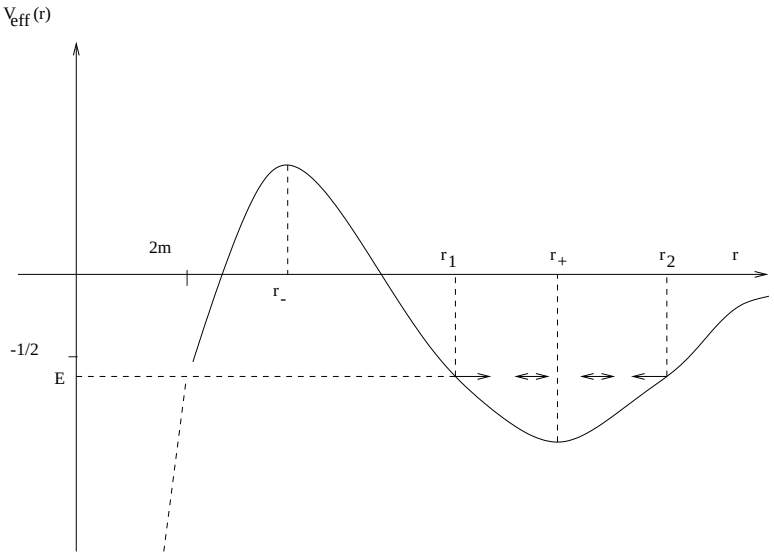


Figure 13: Effective potential for a massive particle with $L/m > \sqrt{12}$. Shown are the maximum of the potential at r_- (an unstable circular orbit), the minimum at r_+ (a stable circular orbit), and the orbit of a particle with $E_{eff} < 0$ with turning points r_1 and r_2.

$E_{eff} = V_{eff}(r_{1,2})$. We will take a closer look at these bound (but not closed) orbits, with their characteristic precession of the perihelion, in section 25.8.

25.7 Some Comments on Circular Timelike Orbits

Recall from above that, provided that $(L/m) > \sqrt{12}$, for a given L there are 2 circular orbits, at the critical points $r_\pm$ of the potential, given by (25.71)

$$r_\pm = (L^2/2m)[1 \pm \sqrt{1 - 12(m/L)^2}] \ , \tag{25.72}$$

For $L \to \sqrt{12}m$ these two circular orbits approach each other, the critical radius tending to $r_\pm \to 6m$. Thus the innermost stable circular orbit (known affectionately as the ISCO in astrophysics) is located at

$$r_{ISCO} = 3r_s = 6m \ . \tag{25.73}$$

This is of course only a relevant quantity when r_{ISCO} lies outside the star, i.e. for $r_0 < 3r_s$ (and thus mainly for black holes or possibly other quite extreme objects like extremely dense neutron stars).

On the other hand, for very large values of L the critical radii are (expand the square root to first order) to be found at

$$(r_+, r_-) \overset{L\to\infty}{\longrightarrow} (L^2/m, 3m) \ . \tag{25.74}$$

Thus letting L range over $[\sqrt{12}m, \infty)$, one sees that circular orbits can occur at any radius $r > 3m$ (for a suitable finetuned choice of angular momentum L). However, such an orbit is stable only for $r > 6m$.

REMARKS:

1. Another way to understand the appearance and significance of the minimal circular radius at $r = 3m$ is to look directly at the radial geodesic equation (25.42)

$$\ddot{r} = -\frac{m}{r^2} + (r - 3m)\frac{L^2}{r^4} \quad . \tag{25.75}$$

   We see that at $r = 3m$ the "centrifugal" term changes sign, and that we can only compensate the first (Newtonian) term on the right-hand side by this term for $r > 3m$.

   We will see in section 25.10 that $r = 3m$ also has a special (and more prominent) significance for *null* geodesics, namely as the location of (unstable) circular null geodesics (the *photon sphere*).

2. From (25.20) one finds that the angular coordinate velocity on a circular orbit $r_\pm$ is

$$\omega_\pm = d\phi/dt = \dot{\phi}/\dot{t} = (L/E)(1 - 2m/r_\pm)/r_\pm^2 \quad . \tag{25.76}$$

   Since this frequency is constant, this immediately gives $\phi = \phi(t)$,

$$\phi(t) = \omega_\pm t + \phi_0 \quad . \tag{25.77}$$

   However, since the quantities $m, E, L, r_\pm$ are not independent of each other, this is not a particularly useful way of writing $\omega_\pm$. First of all, from the effective potential equation and $\dot{r}_\pm = 0$ one finds that the energy E can be written in terms of the other parameters as

$$E^2 = 1 + 2V_{eff}(r_\pm) = (1 - 2m/r_\pm)(1 + L^2/r_\pm^2) \quad . \tag{25.78}$$

   Similarly, from (25.71) L^2 can be written in terms of m and $r_\pm$ as

$$L^2 = mr_\pm^2/(r_\pm - 3m) \quad . \tag{25.79}$$

   Thus one finds that

$$\frac{L^2}{E^2} = \frac{mr_\pm^3}{(r_\pm - 2m)^2} \quad , \tag{25.80}$$

   and the angular velocity (as a function of m and $r_\pm$) is

$$(\omega_\pm)^2 = \frac{m}{r_\pm^3} \quad \Leftrightarrow \quad \omega_\pm^2 r_\pm^3 = m \quad . \tag{25.81}$$

   In particular, for the ISCO one has the characteristic orbital frequency

$$m\omega_{ISCO} = 6^{-3/2} \quad . \tag{25.82}$$

Since ω is inversely proportional to the period of the motion, (25.81) looks very much like Kepler's law which says that the period-squared is proportional to the radius-cubed of an orbit. However, this is a bit of a fluke because this relation is not coordinate-independent but relies on having expressed time and radius in terms of the Schwarzschild coordinates (rather than in terms of proper time along the orbit, say).

25.8 Anomalous Precession of the Perihelia of the Planetary Orbits

Because of the general relativistic correction $\sim 1/r^3$, the bound orbits will not be closed (elliptical). In particular, the position of the perihelion, the point of closest approach of the planet to the sun where the planet has distance r_1, will not remain constant. However, because r_1 is constant, and the planetary orbit is planar, this point will move on a circle of radius r_1 around the sun.

As described in section 25.5, in order to calculate this perihelion shift one needs to calculate the total angle $\Delta\phi$ swept out by the planet during one revolution by integrating this from r_1 to r_2 and back again to r_1, or

$$\Delta\phi = 2\int_{r_1}^{r_2} \frac{d\phi}{dr} dr \quad . \tag{25.83}$$

Rather than trying to evaluate the above integral via some sorcery, we will determine $\Delta\phi$ by analysing the orbit equation (25.64) for $\epsilon = -1$,

$$u'' + u = \frac{m}{L^2} + 3mu^2 \quad . \tag{25.84}$$

In the Newtonian approximation, this equation reduces to that of a displaced harmonic oscillator,

$$u_0'' + u_0 = \frac{m}{L^2} \quad \Leftrightarrow \quad (u_0 - m/L^2)'' + (u_0 - m/L^2) = 0 \quad , \tag{25.85}$$

and the solution is a Kepler ellipse described parametrically by

$$u_0(\phi) = \frac{m}{L^2}(1 + e\cos\phi) \tag{25.86}$$

where e is the eccentricity ($e = 0$ means constant radius and hence a circular orbit). Plugging this back into the Newtonian non-linear 1st-order equation (cf. (25.61))

$$u_0'^2 + u_0^2 = \frac{E^2 - 1}{L^2} + \frac{2m}{L^2}u_0 \quad , \tag{25.87}$$

one finds that the integration constant e is related to the energy by

$$e^2 = 1 + \frac{L^2}{m^2}(E^2 - 1) = 1 + \frac{2L^2}{m^2}E_{eff} \quad . \tag{25.88}$$

In particular, $e^2 < 1$ for bound states (bounded orbits) with $E_{eff} < 0$, and we will concentrate on these orbits. The perihelion (aphelion) is then at $\phi = 0$ ($\phi = \pi$), with

$$r_{1,2} = \frac{L^2}{m}\frac{1}{1 \pm e} \quad . \tag{25.89}$$

Thus the semi-major axis a of the ellipse,

$$2a = r_1 + r_2 \ , \tag{25.90}$$

is

$$a = \frac{L^2}{m}\frac{1}{1-e^2} \ . \tag{25.91}$$

In particular, in the Newtonian theory, one has

$$(\Delta\phi)_0 = 2\int_0^\pi d\phi = 2\pi \ . \tag{25.92}$$

The anomalous perihelion shift due to the effects of General Relativity is thus

$$\delta\phi = \Delta\phi - 2\pi \ . \tag{25.93}$$

In order to determine $\delta\phi$, we now seek a solution to (25.84) of the form

$$u = u_0 + u_1 \tag{25.94}$$

where u_1 is a small deviation. This leads to the equation

$$u_1'' + u_1 = 3mu_0^2 \ . \tag{25.95}$$

The general solution of this inhomogenous differential equation is the general solution of the homogeneous equation (we are not interested in) plus a special solution of the inhomogeneous equation. Noting that this is a forced harmonic oscillator equation, with the frequency of the force commensurate with the frequency of the oscillator, we expect to encounter a resonance phenomenon leading to a non-periodic (and linearly growing) contribution to the solution (just as a special solution of $\ddot{x}(t) + x(t) = \sin t$ is $x(t) = (t/2)\sin t$).

This expectation is indeed borne out. Writing

$$(1 + e\cos\phi)^2 = (1 + \tfrac{1}{2}e^2) + 2e\cos\phi + \tfrac{1}{2}e^2\cos 2\phi \tag{25.96}$$

and noting that

$$\begin{aligned} (\phi\sin\phi)'' + \phi\sin\phi &= 2\cos\phi \\ (\cos 2\phi)'' + \cos 2\phi &= -3\cos 2\phi \end{aligned} \tag{25.97}$$

one sees that a special solution is

$$u_1(\phi) = \frac{3m^3}{L^4}((1 + \tfrac{1}{2}e^2) - \tfrac{1}{6}e^2\cos 2\phi + e\phi\sin\phi) \ . \tag{25.98}$$

The term of interest to us is the third (non-periodic, resonance) term which provides a cumulative non-periodic effect over successive orbits. Focussing on this term, we can write the approximate solution to the orbit equation as

$$u(\phi) \approx \frac{m}{L^2}(1 + e\cos\phi + \frac{3m^2e}{L^2}\phi\sin\phi) \ . \tag{25.99}$$

If the first perihelion is at $\phi = 0$, the next one will be at a point $\Delta\phi = 2\pi + \delta\phi$ close to 2π which is such that $u'(\Delta\phi) = 0$ or

$$\sin\delta\phi = \frac{3m^2}{L^2}(\sin\delta\phi + (2\pi + \delta\phi)\cos\delta\phi) \quad . \tag{25.100}$$

Using that $\delta\phi$ is small, and keeping only the lowest order terms in this equation, one finds the result

$$\delta\phi = \frac{6\pi m^2}{L^2} = 6\pi(\frac{G_N M}{cL})^2 \quad . \tag{25.101}$$

An alternative way to obtain this result is to observe that (25.99) can be approximately written as

$$u(\phi) \approx \frac{m}{L^2}\left(1 + e\cos[(1 - \frac{3m^2}{L^2})\phi]\right) \tag{25.102}$$

From this equation it is manifest that during each orbit the perihelion advances by

$$\delta\phi = 2\pi\frac{3m^2}{L^2} \tag{25.103}$$

$(2\pi(1 - 3m^2/L^2)(1 + 3m^2/L^2) \approx 2\pi)$ in agreement with the above result. In terms of the eccentricity e and the semi-major axis a (25.91) of the elliptical orbit, this can be written as

$$\delta\phi = \frac{6\pi G_N}{c^2}\frac{M}{a(1 - e^2)} \quad . \tag{25.104}$$

Thus general relativity predicts a deviation from the Kepler orbits of the planets, manifesting itself in a precession of the perihelia. As the parameters entering (25.104) are known for the planetary orbits, $\delta\phi$ can be evaluated. Can this also be observed and tested?[78]

At first sight, this seems difficult. Even for Mercury, where this effect is largest (because it has the largest eccentricity) one only finds a $\delta\phi$ of the order of $0,1''$ per revolution. This is of course a tiny effect (1 second, $1''$, is one degree divided by 3600) and not *per se* detectable. However,

1. this effect is cumulative, i.e. after N revolutions one has an anomalous perihelion shift $N\delta\phi$;
2. Mercury has a very short solar year, with about 415 revolutions per century;
3. and accurate observations of the orbit of Mercury go back over more than 200 years.

Thus the cumulative effect is approximately 10^3 $\delta\phi$ and this *is* sufficiently large to be observable in principle. And indeed such an effect is observed (and had for a long time presented a puzzle, an anomaly, for astronomers).

[78] Since I have no independent way of checking them, the following facts and numbers are taken from S. Weinberg, *Gravitation and Cosmology*, section 8.6.

In actual fact, the perihelion of Mercury's orbit, as observed from the Earth, shows a rather significant precession rate of $5601''$ per century, which seems to flatly contradict the Newtonian result that there should be no such precession at all.

However,

- about $5025''$ are due to fact that one is using a non-inertial geocentric coordinate system (precession of the equinoxes)
- $532''$ are due to perturbations of Mercury's orbit caused by the (Newtonian) gravitational attraction of the other planets of the solar system (chiefly Venus, earth and Jupiter).

This much was known prior to General Relativity and left an unexplained anomalous perihelion shift of

$$\delta\phi_{\text{anomalous}} = 43,11'' \pm 0,45''/\text{century} \quad . \tag{25.105}$$

Various ad hoc explanations for this tiny discrepancy between theory and observation (like postulating an additional planet, called *Vulcan*, on an orbit closer to the sun than that of Mercury) were proposed and dismissed.

From (25.104) the prediction of General Relativity for the precession of the perihelion due the the general relativistic correction to the Newtonian theory can be calculated to be

$$\delta\phi_{\text{GR}} = 43,03''/\text{century} \quad . \tag{25.106}$$

Thus general relativity, with its general relativistic correction to the (unperturbed) Kepler orbit of Mercury, appears to account precisely for the observed "anomaly". This is quite a striking, remarkable and impressive confirmation of a prediction of general relativity.

Other observations, involving e.g. the mini-planet Icarus, discovered in 1949, with a huge eccentricity $e \sim 0,827$, binary pulsar systems, and more recently obervations of highly eccentric stars close to the galactic (black hole) center have provided further confirmation of the agreement between General Relativity and observations.

25.9 Null Geodesics

To study the behaviour of massless particles (photons) in the Schwarzschild geometry, we need to study the effective potential

$$V_{eff}(r) = \frac{L^2}{2r^2} - \frac{mL^2}{r^3} = \frac{L^2}{2r^2}(1 - \frac{2m}{r}) \quad . \tag{25.107}$$

The following properties are immediate:

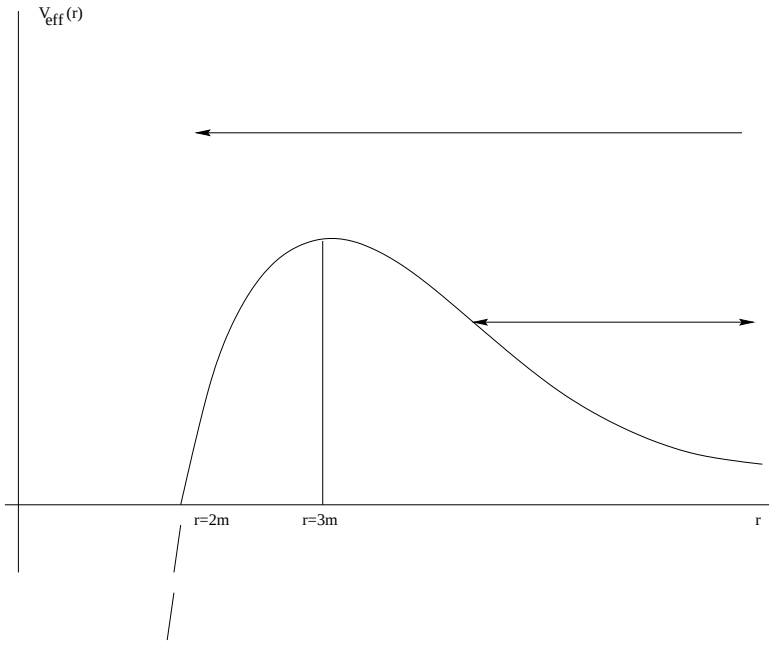


Figure 14: Effective potential for a massless particle. Displayed is the location of the unstable circular orbit at $r = 3m$. A photon with an energy $E^2 < L^2/27m^2$ will be deflected (lower arrow), photons with $E^2 > L^2/27m^2$ will be captured by the star.

1. For $r > 2m$, the potential is positive, $V(r) > 0$.
2. For $r \to \infty$, one has $V_{eff}(r) \to 0$.
3. $V_{eff}(r = 2m) = 0$.
4. When $L = 0$, the photons feel no potential at all.
5. There is one critical point of the potential, at $r = 3m$, with $V_{eff}(r = 3m) = L^2/54m^2$.

Thus the potential has the form sketched in Figure 14, with the following consequences:

1. For energies $E^2 > L^2/27m^2$, photons are captured by the star and will spiral into it. For energies $E^2 < L^2/27m^2$, on the other hand, there will be a turning point, and lightrays will be deflected by the star.

   As this may sound a bit counterintuitive (shouldn't a photon with higher energy be more likely to zoom by the star without being forced to spiral into it?), think about this in the following way. $L = 0$ corresponds to a photon falling radially towards the star, L small corresponds to a slight deviation from radial motion, while L large (thus $\dot{\phi}$ large) means that the photon is travelling along a trajectory that will not bring it very close to the star at all (see the next subsection for the

precise relation between the angular momentum L and the impact parameter b of the photon). It is then not surprising that photons with small L are more likely to be captured by the star (this happens for $L^2 < 27m^2E^2$) than photons with large L which will only be deflected in their path. We will study this in more detail below.

2. Now let us also consider the opposite situation, that of light from or near the star (and we are of course assuming that $r_0 > r_s$). Then for $r_0 < 3m$ and $E^2 < L^2/27m^2$, the light cannot escape to infinity but falls back to the star, whereas for $E^2 > L^2/27m^2$ light will escape. Thus for a path sufficiently close to radial (L small, because $\dot{\phi}$ is then small) light can always escape as long as $r_0 > 2m$.

3. Finally, there is one critical point of the potential, at $r = 3m$. We will (briefly) discuss this case separately below.

25.10 Some Comments on Circular Null Orbits (The Photon Sphere)

As noted above, the effective potential has one critical point, a maximum, at $r = 3m$. This can also be seen from the equation of motion (25.44)

$$\ddot{r} = \frac{L^2}{r^4}(r - 3m) \quad . \tag{25.108}$$

Thus there exists one unstable circular orbit for photons at $r = 3m$ (more precisely one in each "equatorial plane").

While these orbits have some properties in common with the circular timelike orbits discussed in section 25.7, there are also some interesting differences. The first and crucial difference is that this circular orbit can only arise at one particular value of r, namely at $r = 3m$, while (as we have seen) circular timelike orbits can exist for any $r > 3m$. Here are some more brief comments and observations.

Remarks:

1. While not relevant for the applications to the solar system in this section, where we are dealing with objects with a size much larger than $3m$, the existence of this sphere of unstable photon orbits, the so-called *photon sphere*, turns out to be of some interest in black hole astrophysics (as a possibly observable signature of black holes or other very compact objects).

2. Since

$$r = 3m \quad \Rightarrow \quad \dot{t} = E/f(r = 3m) = 3E \quad , \quad \dot{\phi} = L/(3m)^2 = L/(9m^2) \quad , \tag{25.109}$$

these circular null geodesics are characterised by the null wave vector

$$(k^\alpha) = (\dot{t}, \dot{r}, \dot{\phi}, \dot{\theta}) = (3E, 0, L/9m^2, 0) \ , \tag{25.110}$$

with E and L related either by the null condition or, of course equivalently, by the effective potential equation,

$$\dot{r}|_{r=3m} = 0 \quad \Rightarrow \quad E^2 = 2V_{eff}(r = 3m) = L^2/(27m^2) \ . \tag{25.111}$$

Thus the wave vector can e.g. be written as

$$(k^\alpha) = (L/\sqrt{3}m, 0, L/9m^2, 0) \ . \tag{25.112}$$

Of course, the overall normalisation is irrelevant (as it depends on the choice of affine parameter).

3. In this case, the angular velocity (frequency) is given by

$$\omega = d\phi/dt = \dot{\phi}/\dot{t} = (L/E)(1 - 2m/r)/r^2|_{r=3m} = L/(27Em^2) \ . \tag{25.113}$$

Using the relation (25.111), this can be written as

$$\omega = E/L = 1/(3\sqrt{3}m) \quad \Leftrightarrow \quad m\omega = 3^{-3/2} \tag{25.114}$$

This agrees with the $r_- \to 3m$ ($L \to \infty$) limit of the frequency ω_- of the unstable circular timelike orbit (25.81) at $r = r_-$,

$$\lim_{r_- \to 3m} (m\omega_-) = 3^{-3/2} \tag{25.115}$$

(and is structurally similar to, but different from, the ISCO frequency $m\omega_{ISCO} = 6^{-3/2}$ (25.82)).

25.11 Bending of Light by a Star: 3 Derivations

To study the bending of light by a star, we consider an incoming photon (or lightray) with impact parameter b (see Figure 15) and we need to calculate $\phi(r)$ for a trajectory with turning point at $r = r_1$. At that point we have $\dot{r} = 0$. Here the dot can, as usual, be taken to be the derivative with repect to some affine parameter σ. However, noting that the condition $g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$ is reparametrisation-invariant (unlike its massive cousin $g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = -1$), we can equally well choose to parametrise the lightrays by the coordinate time t even though this is not an affine parameter (this matters at the level of the 2nd order geodesic differential equations but not at the level of the 1st integrals and the effective potential).

Either way r_1 is determined by

$$E_{eff} = V_{eff}(r_1) \quad \Leftrightarrow \quad r_1^2 = \frac{L^2}{E^2}(1 - \frac{2m}{r_1}) \ . \tag{25.116}$$

The first thing we need to establish is the relation between b and the other parameters E and L. Consider the ratio

$$\frac{L}{E} = \frac{r^2\dot{\phi}}{(1-2m/r)\dot{t}} \quad . \tag{25.117}$$

For large values of r, $r \gg 2m$, this reduces to

$$\frac{L}{E} = r^2\frac{d\phi}{dt} \quad . \tag{25.118}$$

On the other hand, for large r we can approximate $b/r = \sin\phi$ by ϕ. Since we also have $dr/dt = -1$ (for an incoming lightray), we deduce

$$\frac{L}{E} = r^2\frac{d}{dt}\frac{b}{r} = b \quad . \tag{25.119}$$

In terms of the variable $u = 1/r$ the equation for the shape of the orbit (25.64) is

$$u'' + u = 3mu^2 \tag{25.120}$$

and the elliptic integral (25.62) for $\Delta\phi$ is

$$\Delta\phi = 2\int_{r_1}^{\infty}\frac{d\phi}{dr}dr = 2\int_0^{u_1} du\,[b^{-2} - u^2 + 2mu^3]^{-1/2} \quad . \tag{25.121}$$

Moreover, in terms of u we can write the equation (25.116) for $u_1 = 1/r_1$ as

$$b^{-2} = u_1^2 - 2mu_1^3 \quad . \tag{25.122}$$

In the absence of the general relativistic correction (calling this 'Newtonian' is perhaps not really appropriate since we are dealing with photons/lightrays) one has $b^{-1} = u_1$ or $b = r_1$ (no deflection). The orbit equation

$$u_0'' + u_0 = 0 \tag{25.123}$$

has the solution

$$u_0(\phi) = \frac{1}{b}\sin\phi \quad , \tag{25.124}$$

describing the straight line

$$r_0(\phi)\sin\phi = b \quad . \tag{25.125}$$

Obligingly the integral gives

$$(\Delta\phi)_0 = 2\int_0^1 dx(1-x^2)^{-1/2} = 2\arcsin 1 = \pi \quad . \tag{25.126}$$

Thus the deflection angle is related to $\Delta\phi$ by

$$\delta\phi = \Delta\phi - \pi \quad . \tag{25.127}$$

We will now determine $\delta\phi$ in three different ways,

- by perturbatively solving the orbit equation (25.120);
- by perturbatively evaluating the elliptic integral (25.121);
- by performing a perturbative expansion (linearisation) of the Schwarzschild metric.

Derivation I: Perturbative Solution of the Orbit Equation

In order to solve the orbit equation (25.120), we proceed as in section 25.8. Thus the equation for the (small) deviation $u_1(\phi)$ is

$$u_1'' + u_1 = 3mu_0^2 = \frac{3m}{b^2}(1 - \cos^2\phi) = \frac{3m}{2b^2}(1 - \cos 2\phi) \tag{25.128}$$

which has the particular solution (cf. (25.97))

$$u_1(\phi) = \frac{3m}{2b^2}(1 + \tfrac{1}{3}\cos 2\phi) \quad . \tag{25.129}$$

Therefore

$$u(\phi) = \frac{1}{b}\sin\phi + \frac{3m}{2b^2}(1 + \tfrac{1}{3}\cos 2\phi) \quad . \tag{25.130}$$

By considering the behaviour of this equation as $r \to \infty$ or $u \to 0$, one finds an equation for (minus) half the deflection angle, namely

$$\frac{1}{b}(-\delta\phi/2) + \frac{3m}{2b^2}\frac{4}{3} = 0 \quad , \tag{25.131}$$

leading to the result

$$\delta\phi = \frac{4m}{b} = \frac{4MG_N}{bc^2} \quad . \tag{25.132}$$

Derivation II: Perturbative Evaluation of the Elliptic Integral

The perturbative evaluation of (25.121) is rather tricky when it is regarded as a function of the independent variables m and b, with r_1 determined by (25.116) (try this!). The trick to evaluate (25.121) is (see R. Wald, *General Relativity*) to regard the integral as a function of the independent variables r_1 and m, with b eliminated via (25.122). Thus (25.121) becomes

$$\Delta\phi = 2\int_0^{u_1} du\, [u_1^2 - u^2 - 2m(u_1^3 - u^3)]^{-1/2} \quad . \tag{25.133}$$

The first order correction

$$\Delta\phi = (\Delta\phi)_0 + m(\Delta\phi)_1 + \mathcal{O}(m^2) \tag{25.134}$$

is therefore

$$(\Delta\phi)_1 = \left(\frac{\partial}{\partial m}\Delta\phi\right)_{m=0} = 2\int_0^{b^{-1}} du\, \frac{b^{-3} - u^3}{(b^{-2} - u^2)^{3/2}} \quad . \tag{25.135}$$

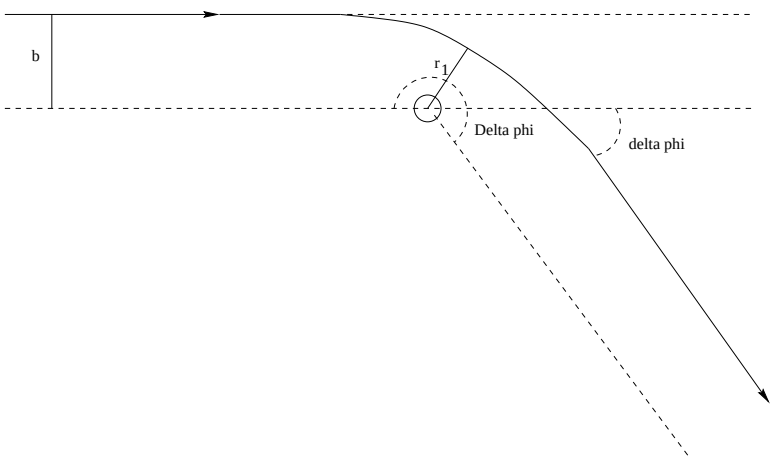


Figure 15: Bending of light by a star. Indicated are the definitions of the impact parameter b, the perihelion r_1, and of the angles $\Delta\phi$ and $\delta\phi$.

This integral is elementary,

$$\int dx\, \frac{1-x^3}{(1-x^2)^{3/2}} = -(x+2)\left(\frac{1-x}{1+x}\right)^{1/2} \quad , \tag{25.136}$$

and thus

$$(\Delta\phi)_1 = 4b^{-1} \quad , \tag{25.137}$$

leading to

$$\delta\phi = \frac{4m}{b} = \frac{4MG_N}{bc^2} \quad , \tag{25.138}$$

in agreement with the result (25.132) obtained above.

Derivation III: Linearising the Schwarzschild Metric

It is instructive to look at the second derivation from another point of view. As we will see, in some sense this derivation 'works' because the bending of light is accurately described by the *linearised solution*, i.e. by the metric that one obtains from the Schwarzschild metric by the approximation

$$\begin{aligned} A(r) = 1 - \frac{2m}{r} \quad &\to \quad 1 - \frac{2m}{r} \\ B(r) = (1 - \frac{2m}{r})^{-1} \quad &\to \quad 1 + \frac{2m}{r} \quad . \end{aligned} \tag{25.139}$$

I will only sketch the main steps in this calculation, so you should think of this subsection as an annotated exercise.

First of all, redoing the analysis of sections 25.3 and 25.5 for a general spherically symmetric static metric (24.6),

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad , \tag{25.140}$$

it is easy to see that the orbit equation can be written as

$$B(r)\frac{r'^2}{r^4} + \frac{1}{r^2} = \frac{\epsilon}{L^2} + \frac{E^2}{L^2 A(r)} \tag{25.141}$$

or, in terms of $u = 1/r$, as (abusing notation by writing $A(r = 1/u)$ as $A(u)$ etc., apologies if this causes allergic reactions)

$$B(u)\ u'^2 + u^2 = \frac{\epsilon}{L^2} + \frac{E^2}{L^2 A(u)} \quad . \tag{25.142}$$

We will concentrate on the lightlike case $\epsilon = 0$,

$$B(u)\ u'^2 + u^2 = \frac{E^2}{L^2 A(u)} \quad , \tag{25.143}$$

and express the impact parameter $b = L/E$ in terms of the turning point $r_1 = 1/u_1$ of the trajectory. At this turning point, $u' = 0$, and thus

$$\frac{E^2}{L^2} = A(u_1)u_1^2 \quad , \tag{25.144}$$

leading to

$$B(u)\ u'^2 = \frac{A(u_1)}{A(u)}u_1^2 - u^2 \quad . \tag{25.145}$$

We thus find

$$\frac{d\phi}{du} = \pm B(u)^{1/2}\left[\frac{A(u_1)}{A(u)}u_1^2 - u^2\right]^{-1/2} \quad . \tag{25.146}$$

For the (linearised) Schwarzschild metric the term in square brackets is

$$\begin{aligned} \frac{A(u_1)}{A(u)}u_1^2 - u^2 &= u_1^2(1 + 2m(u - u_1)) - u^2 \\ &= (u_1^2 - u^2)(1 - 2m\frac{u_1^2}{u_1 + u}) \quad . \end{aligned} \tag{25.147}$$

Using this and the approximate (linearised) value for $B(u)$,

$$B(u) \approx 1 + 2mu \tag{25.148}$$

one finds that $d\phi/du$ is given by

$$\begin{aligned} \frac{d\phi}{du} &= \pm B(u)^{1/2}\left[(u_1^2 - u^2)(1 - 2m\frac{u_1^2}{u_1 + u})\right]^{-1/2} \\ &\approx (u_1^2 - u^2)^{-1/2}\left(1 + m(\frac{u_1^2}{u_1 + u} + u)\right) \\ &= (u_1^2 - u^2)^{-1/2} + m\frac{u_1^3 - u^3}{(u_1^2 - u^2)^{3/2}} \quad . \end{aligned} \tag{25.149}$$

The first term now gives us the Newtonian result and, comparing with Derivation II, we see that the second term agrees precisely with the integrand of (25.135) with $b \to r_1$

(which, in a term that is already of order m, makes no difference). We thus conclude that the deflection angle is, as before,

$$\delta\phi = 2\int_0^{u_1} du\; m\frac{u_1^3 - u^3}{(u_1^2 - u^2)^{3/2}} = 4mu_1 \approx \frac{4m}{b} \quad . \tag{25.150}$$

REMARKS:

1. This effect is physically measurable and was one of the first true tests of Einstein's new theory of gravity. For light just passing the sun the predicted value is

   $$\delta\phi \sim 1,75'' \quad . \tag{25.151}$$

   Quick sanity check on this estimate: $1'' = \pi/(180 \times 3600) \approx 0.5 \times 10^{-5}$ radians, while using the (rough) values for $r_s = 2m$ and $r_0 \approx b$ for the sun given in (24.40), one has

   $$\frac{4m}{b} \approx (6/7) \times 10^{-5} \approx (12/7)'' \approx 1.7'' \quad . \tag{25.152}$$

   Experimentally this is a bit tricky to observe because one needs to look at light from distant stars passing close to the sun. Under ordinary circumstances this would not be observable, but in 1919 a test of this was performed during a total solar eclipse, by observing the effect of the sun on the apparent position of stars in the direction of the sun. The observed value was rather imprecise, yielding $1,5'' < \delta\phi < 2,2''$ which is, if not a confirmation of, at least consistent with General Relativity.

2. More recently, it has also been possible to measure the deflection of radio waves by the gravitational field of the sun. These measurements rely on the fact that a particular Quasar, known as 3C275, is obscured annually by the sun on October 8th, and the observed result (after correcting for diffraction effects by the corona of the sun) in this case is $\delta = 1,76'' \pm 0,02''$.

3. The value predicted by General Relativity is, interestingly enough, exactly twice the value that would have been predicted by the Newtonian approximation of the geodesic equation alone (but the Newtonian approximation is not valid anyway because it applies to slowly moving objects, and light certainly fails to satisfy this condition). A calculation leading to this wrong value had first been performed by Soldner in 1801 (!) (by cancelling the mass m out of the Newtonian equations of motion before setting $m = 0$) and also Einstein predicted this wrong result in 1908 (his equivalence principle days, long before he came close to discovering the field equations of General Relativity now carrying his name).

   This result can be obtained from the above calculation by setting $B(u) = 1$ instead of (25.148), as in the Newtonian approximation only g_{00} is non-trivial.

4. More generally, one can calculate the deflection angle for a metric with the approximate behaviour

$$B(u) \approx 1 + 2\gamma m u \ , \tag{25.153}$$

for γ a real parameter, with the result

$$\delta\phi \approx \frac{1+\gamma}{2}\,\frac{4m}{b} \ . \tag{25.154}$$

This reproduces the previous result for $\gamma = 1$, half its value for $\gamma = 0$, and checking to which extent measured deflection angles agree with the theoretical prediction of general relativity ($\gamma = 1$) constitutes an experimental test of general relativity. In this context γ is known as one of the PPN parameters (PPN for *parametrised post-Newtonian approximation*).

25.12 A Unified Description in terms of the Runge-Lenz Vector

The perhaps slickest way to obtain the orbits of the Kepler problem is to make use of the so-called *Runge-Lenz vector* (even though it was discovered neither by Runge nor by Lenz).[79]

Recall that, due to conservation of angular momentum $\vec{L}$, the orbits in any spherically symmetric potential are planar. The bound orbits of the Kepler problem, however, have the additional property that they are closed, i.e. that the perihelion is constant. This suggests that there is a further hidden symmetry in the Kepler problem, with the position of the perihelion the corresponding conserved charge. This is indeed the case.

Consider, for a spherically symmetric potential $W(r)$, the vector

$$\vec{A} = \dot{\vec{x}} \times \vec{L} + W(r)\vec{x} \tag{25.155}$$

or, in components,

$$A_i = \epsilon_{ijk}\dot{x}_j L_k + W(r)x_i \ . \tag{25.156}$$

A straightforward calculation, using the Newtonian equations of motion in the potential $W(r)$, shows that

$$\frac{d}{dt}A_i = (r\partial_r W(r) + W(r))\dot{x}_i \ . \tag{25.157}$$

Thus $\vec{A}$ is conserved if and only if $W(r)$ is homogeneous of degree (-1),

$$\frac{d}{dt}\vec{A} = 0 \ \Leftrightarrow \ W(r) = \frac{c}{r} \ . \tag{25.158}$$

In our notation, $c = \epsilon m$, and we will henceforth refer to the vector

$$\vec{A} = \dot{\vec{x}} \times \vec{L} + \frac{\epsilon m}{r}\vec{x} \tag{25.159}$$

[79] For an account of the history, with further references, see e.g. the Wikipedia entry `https://en.wikipedia.org/wiki/Laplace-Runge-Lenz_vector`.

as the Runge-Lenz vector.

In addition to being preserved for a $(1/r)$-potential, the Runge-Lenz vector has the following properties:

1. $\vec{A}$ is orthogonal to $\vec{L}$ and hence lies in the plane of the orbit,

$$\vec{x}.\vec{L} = 0 \quad \Rightarrow \quad \vec{A}.\vec{L} = 0 \ . \tag{25.160}$$

2. The norm A of $\vec{A}$ can be expressed in terms of the other conserved quantities and parameters (energy E, angular momentum L, mass m) of the problem. In the notation of section 25.3 one has

$$A^2 = E^2 L^2 + \epsilon(L^2 + \epsilon m^2) \ . \tag{25.161}$$

3. Thus, even though a priori $\vec{A}$ has 3 components, the only new information is contained in the constant direction of $\vec{A}$, which (since it lies in the orbital plane) is just one angle, a single real number. Thus all in all, in the Kepler problem one has 5 independent constants of motion, $E, \vec{L}$ and the direction of $\vec{A}$.

4. It is well known, and can be shown e.g. by determining the Poisson brackets among the L_i and A_j (the calculation of $\{A_i, A_j\}$ is a bit messy), and a suitable rescaling of the A_i by a function of the (conserved) energy, that $\vec{A}$ extends the manifest symmetry group of rotations $SO(3)$ of the Kepler problem to the (hidden, phase space) symmetry group $SO(4)$ for bound orbits and $SO(3,1)$ for scattering orbits. We will not need to make use of this fact here, though.

Given all this information, it is now straightforward to determine the Keplerian orbits from $\vec{A}$. Let us choose the constant direction of $\vec{A}$ to be in the direction $\phi = 0$. Then $\vec{A}.\vec{x} = Ar\cos\phi$ and from (25.159) one finds

$$Ar\cos\phi = L^2 + \epsilon m r \ . \tag{25.162}$$

Now we consider the two cases $\epsilon = -1$ and $\epsilon = 0$.

For $\epsilon = -1$, (25.162) can be written as

$$\frac{1}{r(\phi)} = \frac{m}{L^2}(1 + \frac{A}{m}\cos\phi) \ . \tag{25.163}$$

Comparing with (25.86), we recognise this as the equation for an ellipse with eccentricity e and semi-major axis a (25.91) given by

$$e = \frac{A}{m} \qquad \frac{m}{L^2} = \frac{1}{a(1-e^2)} \ . \tag{25.164}$$

Moreover, we see that the perihelion is at $\phi = 0$ which establishes that the Runge-Lenz vector points from the center of attraction to the (constant) position of the perihelion. During one revolution the angle ϕ changes from 0 to 2π.

For $\epsilon = 0$ (i.e. no potential), on the other hand, (25.162) reduces to

$$\frac{1}{r(\phi)} = \frac{A}{L^2}\cos\phi \tag{25.165}$$

This describes a straight line (25.124) with impact parameter

$$b = \frac{L^2}{A} = \frac{L}{E} \ . \tag{25.166}$$

In this case, ϕ runs from $-\pi/2$ to $\pi/2$ and the point of closest approach is again at $\phi = 0$ (distance b).

We see that the Runge-Lenz vector captures precisely the information that in the Newtonian theory bound orbits are closed and lightrays are not deflected. The Runge-Lenz vector will no longer be conserved in the presence of the general relativistic correction to the Newtonian motion, and this non-constancy is a precise measure of the deviation of the general relativistic orbits from their Newtonian counterparts. As shown e.g. in an article by Brill and Goel[80] this provides a very elegant and quick way of (re-)deriving the results about perihelion precession and deflection of light in the solar system.

Calculating the time-derivative of the Newtonian Runge-Lenz $\vec{A}$ (25.159), but now for a particle moving in the general relativistic potential (25.34)

$$V(r) = \epsilon\frac{m}{r} - \frac{mL^2}{r^3} \ , \tag{25.167}$$

one finds one additional term arising from substituting the equation of motion into $\ddot{x}_j$, leading to (of course we now switch from t to τ)

$$\frac{d}{d\tau}\vec{A} = \frac{3mL^2}{r^2}\frac{d}{d\tau}\vec{n} \ , \tag{25.168}$$

where $\vec{n} = \vec{x}/r = (\cos\phi, \sin\phi, 0)$ is the unit vector in the equatorial plane $\theta = \pi/2$ of the orbit. Thus $\vec{A}$ rotates in the ϕ-direction in the equatorial plane. If $\vec{A}$ is originally pointing in the x^1-direction $\phi = 0$, then its initial angular velocity in the x^2-direction is

$$\omega = \frac{3mL^2\cos\phi}{Ar^2}\dot{\phi} \ . \tag{25.169}$$

In principle, here A refers to the norm of the Newtonian Runge-Lenz vector (25.159) calculated for a trajectory $\vec{x}(\tau)$ in the general relativistic potential (25.167). This norm is now no longer constant,

$$A^2 = E^2L^2 + \epsilon(L^2 + \epsilon m^2) + \frac{2mL^4}{r^3} \ . \tag{25.170}$$

[80] D. Brill, D. Goel, *Light bending and perihelion precession: A unified approach*, Am. J. Phys. 67 (1999) 316, `arXiv:gr-qc/9712082`

However, assuming that the change in $\vec{A}$ is small, we obtain an approximate expression for ω by substituting the unperturbed orbit $r_0(\phi)$ from (25.162),

$$r_0(\phi) = \frac{L^2}{A\cos\phi - \epsilon m} \quad , \tag{25.171}$$

as well as the unperturbed norm (25.161) in (25.169) to find

$$\omega \approx \frac{3m}{AL^2}(A\cos\phi - \epsilon m)^2 \cos\phi \, \dot{\phi} \quad . \tag{25.172}$$

Now the total change in the direction of $\vec{A}$ when the object moves from ϕ_1 to ϕ_2 can be calculated from

$$\begin{aligned} \delta\phi &= \int_{\phi_1}^{\phi_2} \omega d\tau \\ &= \frac{3m}{AL^2} \int_{\phi_1}^{\phi_2} d\phi \, (A\cos\phi - \epsilon m)^2 \cos\phi \quad . \end{aligned} \tag{25.173}$$

For $\epsilon = -1$, and $(\phi_1, \phi_2) = (0, 2\pi)$, this results in (only the $\cos^2\phi$-term gives a non-zero contribution)

$$\delta\phi = 2\pi \frac{3m^2}{L^2} = \frac{6\pi m^2}{L^2} \quad , \tag{25.174}$$

in precise agreement with (25.101,25.103).

For $\epsilon = 0$, on the other hand, one has

$$\delta\phi = \frac{3mA}{L^2} \int_{-\pi/2}^{\pi/2} d\phi \, \cos^3\phi \quad . \tag{25.175}$$

Using

$$\int \cos^3\phi = \sin\phi - \tfrac{1}{3}\sin^3\phi \quad , \tag{25.176}$$

one finds

$$\delta\phi = \frac{4mA}{L^2} = \frac{4m}{b} \quad , \tag{25.177}$$

which agrees precisely with the results of section 25.11.